Item Name	Type	Description
CCSDS header	12 bytes	See section 2.1 for details
SCI0AID	U32	See above (same as waveform packets)
SCI0TYPE	U8	
SCI0CONT	U8	
SCI0SPARE1	U13	
SCI0PACK	U1	
SCI0FRAG	U1	
SCI0COMP	U1	
SCI0EVTNUM	U16	
SCI0CAT	U8	
SCI0QUAL	U8	
SCI0FRAGOFF	U16	
SCI0VER	U16	
SCI0TIMECOARSE	U16	
SCI0TIMEFINE	U16	
SCI0SPARE2	U32	
SCI0SPARE3	U32	
SCI0SPARE4	U32	
SCIDATALENGTH	U32	Length of raw science data in memory including this FPGA pre-pended metadata header but excludes the sync words at the start of all Flash pages.
TIMESTAMP1	U16	Padded with 0's
	U16	Time of event trigger, in integer seconds (bits 31:16)
TIMESTAMP2	U16	Time of event trigger, in integer seconds (bits 15:0)
	U16	Time of event trigger, in sub-seconds (20usec per DN) event time, sub-seconds
EVENTNUMBER	U1	Padded with 0
	U3	Trigger offset: number of WCLK cycles that the start of post-trigger data collection is delayed from the trigger position to align the HS and LS sampling periods. See FPGA's ADC Trigger Offset" register.
	U2	Padded with 0's
	U10	Identifies origin of this trigger (multiple concurrent triggers are possible) · upper 4 bits are spares (unused) · bit 5 – external trigger · bit 4 – SW trigger · bit 3 – LS ADC1 trigger (target low range CSA) · bit 2 – HS ADC1Q trigger (mid-gain channel) · bit 1 – HS ADC0Q trigger (low-gain channel) · bit 0 – HS ADC0I trigger (high-gain channel) See FPGA's "ADC Trigger ID" register.
	U16	Event number assigned by FPGA. Resets to 0 at power on and incremented thereafter (rolling over from 0xFFFF to 0).
NBLOCKS, see FPGA's "ADC N Blocks" register	U1	HS Blocks Error - this bit is set if total number of HS blocks (pre + post) is greater than 16
	U1	LS Blocks Error - this bit is set if total number of LS blocks (pre + post) is greater than 64
	U6	Dead Blocks Base - programmed base number of dead blocks (0-63)
	U4	Dead Blocks Shift - programmed number of bit positions to shift Dead Blocks Base value to the left to get the total number of dead blocks (0-15)
	U4	HS Pre Blocks - programmed number of pre-trigger blocks for high-rate ADC data (1-16)
	U4	HS Post Blocks - programmed number of post-trigger blocks for low-rate ADC data (1-64)
	U6	LS Pre Blocks - programmed number of post-trigger blocks for low-rate ADC data (1-64)
	U6	LS Post Blocks - programmed number of post-trigger blocks for low-rate ADC data (1-64)
HSADC0IHDR1, see	U10	ADC0I trigger level - trigger level for ADC0I channel

HSADC0IHDR1, see FPGA's "HS ADC0 Channel I Trigger Control 2" register	U11	ADC0I trigger Nmax12 - max # of samples between pulse 1 and 2 for ADC0I double pulse triggering
	U11	ADC0I trigger Nmin12 - min # of samples between pulse 1 and 2 for ADC0I double pulse triggering
HSADC0IHDR2, see FPGA's "HS ADC0 Channel I Trigger Control 1" register	U8	ADC0I trigger Nmax2 - max # of samples for pulse 2 for ADC0I double pulse mode triggering
	U8	ADC0I trigger Nmin2 - min # of samples for pulse 2 for ADC0I double pulse mode triggering
	U8	ADC0I trigger Nmax1 - max # of samples for pulse 1 for ADC0I single and double pulse triggering
	U8	ADC0I trigger Nmin1 - min # of samples for pulse 1 for ADC0I single and double pulse triggering
HSADC0QHDR1, see FPGA's "HS ADC0 Channel Q Trigger Control 2" register	U10	ADC0Q trigger level - trigger level for ADC0Q channel
	U11	ADC0Q trigger Nmax12 - max # of samples between pulse 1 and 2 for ADC0Q double pulse triggering
	U11	ADC0Q trigger Nmin12 - min # of samples between pulse 1 and 2 for ADC0Q double pulse triggering
HSADC0QHDR2, see FPGA's "HS ADC0 Channel Q Trigger Control 1" register	U8	ADC0Q trigger Nmax2 - max # of samples for pulse 2 for ADC0Q double pulse mode triggering
	U8	ADC0Q trigger Nmin2 - min # of samples for pulse 2 for ADC0Q double pulse mode triggering
	U8	ADC0Q trigger Nmax1 - max # of samples for pulse 1 for ADC0Q single and double pulse triggering
	U8	ADC0Q trigger Nmin1 - min # of samples for pulse 1 for ADC0Q single and double pulse triggering
HSADC1QHDR1, see FPGA's "HS ADC1 Channel Q Trigger Control 2" register	U10	ADC1Q trigger level - trigger level for ADC1Q channel
	U11	ADC1Q trigger Nmax12 - max # of samples between pulse 1 and 2 for ADC1Q double pulse triggering
	U11	ADC1Q trigger Nmin12 - min # of samples between pulse 1 and 2 for ADC1Q double pulse triggering
HSADC1QHDR2, see FPGA's "HS ADC1 Channel Q Trigger Control 1" register	U8	ADC1Q trigger Nmax2 - max # of samples for pulse 2 for ADC1Q double pulse mode triggering
	U8	ADC1Q trigger Nmin2 - min # of samples for pulse 2 for ADC1Q double pulse mode triggering
	U8	ADC1Q trigger Nmax1 - max # of samples for pulse 1 for ADC1Q single and double pulse triggering
	U8	ADC1Q trigger Nmin1 - min # of samples for pulse 1 for ADC1Q single and double pulse triggering
LSADC, see FPGA's LS ADC1 Trigger Control" register	U8	Padded with 0's
	U3	coincidence mode blocks - number of blocks coincidence window is enabled after LS ADC1 trigger
	U1	LS ADC1 trigger polarity - trigger polarity for LS ADC1 (0 = normal, 1 = inverted)
	U12	LS ADC1 trigger level - trigger level for LS ADC1
	U8	LS ADC1 trigger Nmin - min number of samples above/below trigger level for triggering LS ADC1
TRIGGERMODE, see FPGA's "Trigger Mode" Register	U1	HVPS polarity status, "cation" mode (0) or "anion" mode (1)
	U1	Stim pulse enabled (1) or disabled (0)
	U21	Padded with 0's
	U1	External trigger polarity
	U1	LS ADC coincidence mode enable
	U1	LS ADC trigger enable
	U2	ADC1Q trigger mode – same as above
	U2	ADC0Q trigger mode – same as above
	U2	ADC0I trigger mode 00 – no triggering 01 – threshold mode 10 – single pulse mode 11 – double pulse mode
SPARE0	U32	Padded with 0's
SPARE1	U32	Padded with 0's
	U20	Padded with 0's

IONDETAILS0	U12	Ion grid (LS ADC2) 15 ms min, new for IDEX. The minimum value on LS ADC2 during the 15 ms window after the previous event trigger.
IONDETAILS1	U16	Ion grid (LS ADC2) saturation high count, new for IDEX. The number of LS ADC2 samples that were equal to 0x000 during the 15 ms window after the previous event trigger.
	U16	Ion grid (LS ADC2) saturation low count, new for IDEX. The number of LS ADC2 samples that were equal to 0xFFF during the 15 ms window after the previous event trigger.
TOFMAX	U2	Padded with 0's
	U10	ADC1Q max value - max sample value on ADC1Q channel since previous event was captured. See FPGA's "ADC1 DQ Channel MinMax" Register.
	U10	ADC0Q max value - max sample value on ADC0Q channel since previous event was captured. See FPGA's "ADC0 DQ Channel MinMax" Register.
	U10	ADC0I max value - max sample value on ADC0I channel since previous event was captured. . See FPGA's "ADC0 DI Channel MinMax" Register.
TOFMIN	U2	Padded with 0's
	U10	ADC1Q min value - min sample value on ADC1Q channel since previous event was captured. See FPGA's "ADC1 DQ Channel MinMax" Register.
	U10	ADC0Q min value - min sample value on ADC0Q channel since previous event was captured. See FPGA's "ADC0 DQ Channel MinMax" Register.
	U10	ADC0I min value - min sample value on ADC0I channel since previous event was captured. See FPGA's "ADC0 DI Channel MinMax" Register.
LSADC0MINMAX, see FPGA's "LS ADC0 Channel MinMax" Register	U8	Padded with 0's
	U12	LS ADC0 max value - max sample value on LS ADC0 channel since previous event was captured
	U12	LS ADC0 min value - min sample value on LS ADC0 channel since previous event was captured
LSADC1MINMAX, see FPGA's "LS ADC1 Channel MinMax" Register	U8	Padded with 0's
	U12	LS ADC1 max value - max sample value on LS ADC1 channel since previous event was captured
	U12	LS ADC1 min value - min sample value on LS ADC1 channel since previous event was captured
LSADC2MINMAX, see FPGA's "LS ADC2 Channel MinMax" Register	U8	Padded with 0's
	U12	LS ADC2 max value - max sample value on LS ADC2 channel since previous event was captured
	U12	LS ADC2 min value - min sample value on LS ADC2 channel since previous event was captured
TOFTRIGGERDELAY	U2	Padded with 0's
	U10	ADC1Q trigger delay - delay in DCLK of trigger condition through comparison FIFO for ADC1Q
	U10	ADC0Q trigger delay - delay in DCLK of trigger condition through comparison FIFO for ADC0Q
	U10	ADC0I trigger delay - delay in DCLK of trigger condition through comparison FIFO for ADC0I
TOFSAMPLEDELAY	U2	Padded with 0's
	U10	ADC1Q sample delay - delay in samples from actual trigger position in event buffer to expected trigger position for ADC1Q
	U10	ADC0Q sample delay - delay in samples from actual trigger position in event buffer to expected trigger position for ADC0Q
	U10	ADC0I sample delay - delay in samples from actual trigger position in event buffer to expected trigger position for ADC0I
TOFTRANSCOUNT, see FPGA's "Transitions Counts" Register	U2	Padded with 0's
	U10	ADC1Q transition count - Number of transitions across the programmed threshold level for ADC1Q
	U10	ADC0Q transition count - Number of transitions across the programmed threshold level for ADC0Q
	U10	ADC0I transition count - Number of transitions across the programmed threshold level for ADC0I
PROCHKADCCHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "current on the 1V POL"
	U4	Padded with 0's

	U12	Last measurement in raw DN for Processor Board signal "current on the 1.9V POL"
PROCHKADCCHAN23	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "ProcBd Temperature 1" (near TOF/HS ADCs)
	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "ProcBd Temperature 2" (near PWB center)
PROCHKADCCHAN45	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "voltage on 1V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "FPGA Temperature"
PROCHKADCCHAN67	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "voltage on 1.9V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for Processor Board signal "voltage on 3.3V bus"
HVPSHKADCCHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Detector Voltage"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Sensor Voltage"
HVPSHKADCCHAN23	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Target Voltage"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Reflectron Voltage"
HVPSHKADCCHAN45	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "Rejection Voltage"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "current for the HVPS detector"
HVPSHKADCCHAN67	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "positive current for the HVPS sensor"
	U4	Padded with 0's
	U12	Last measurement in raw DN for HVPS Board signal "negative current for the HVPS sensor"
LVPSHKADC0CHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage of +3.3V reference"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage of +3.3V operational reference"
LVPSHKADC0CHAN23	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on -6V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +6V bus"
LVPSHKADC0CHAN45	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +16V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +3.3V bus"
LVPSHKADC0CHAN67	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on -5V bus"
	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "voltage on +5V bus"
LVPSHKADC1CHAN01	U4	Padded with 0's
	U12	Last measurement in raw DN for LVPS Board signal "current on +3.3V bus"
	U4	Padded with 0's

	U12	Last measurement in raw DN for LVPS Board signal “current on 16V bus”
LVPSHKADC1CHAN23	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on +6V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on -6V bus”
LVPSHKADC1CHAN45	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on +5V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on -5V bus”
LVPSHKADC1CHAN67	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on +2.5V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “current on -2.5V bus”
LVPSHKADC2CHAN01	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “primary target temperature”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “redundant target temperature”
LVPSHKADC2CHAN23	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “temperature of Decon-Actuator Board (DAB)”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “temperature of LVPS board”
LVPSHKADC2CHAN45	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “temperature of HVPS board (in EBox)”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “voltage on +2.5V bus”
LVPSHKADC2CHAN67	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “voltage on -2.5V bus”
	U4	Padded with 0’s
	U12	Last measurement in raw DN for LVPS Board signal “Spare, which is not connected to anything”
SPARE	U32	Padded with 0’s
SPARE	U32	Padded with 0’s
FSWHDR00	U32	Section reserved for FSW usage
FSWHDR01	U32	Section reserved for FSW usage
FSWHDR02	U32	Section reserved for FSW usage
FSWHDR03	U32	Section reserved for FSW usage
FSWHDR04	U32	Section reserved for FSW usage
FSWHDR05	U32	Section reserved for FSW usage
FSWHDR06	U32	Section reserved for FSW usage
FSWHDR07	U32	Section reserved for FSW usage
FSWHDR08	U32	Section reserved for FSW usage
FSWHDR09	U32	Section reserved for FSW usage
FSWHDR10	U32	Section reserved for FSW usage
FSWHDR11	U32	Section reserved for FSW usage
FSWHDR12	U32	Section reserved for FSW usage
FSWHDR13	U32	Section reserved for FSW usage
FSWHDR14	U32	Section reserved for FSW usage
FSWHDR15	U32	Section reserved for FSW usage
FPGAVER	U32	FPGA version, see FPGA’s “DataTime Register”
SYNCSCI0PKT	U16	Synchronization marker, fixed 0x3333
CRCSCI0PKT	U16	CRC CCITT-16

Item Name	Type	Description
VERSION	U3	CCSDS version number, fixed 0b000
TYPE	U1	CSDS type, fixed 0b0 for telemetry
SEC_HDR_FLG	U1	CCSDS secondary header flag, fixed 0b1 for presence of secondary header
PKT_APID	U11	CCSDS Application Process Identifier (APID), either 0x590 or 0x591
SEQ_FLGS	U2	CCSDS grouping flags, fixed 0xb11 for standalone packet
SRC_SEQ_CTR	U14	CCSDS source sequence counter, increments per packet (separate counters per APID)
PKT_LEN	U16	CCSDS packet length, excludes UMS and CCSDS primary header (minus one)
SHCOARSE	U32	Time of packet generation (not data acquisition), as integer seconds since epoch
SHFINE	U16	Time of packet generation, each DN represents 20usec within current second
SCI0AID	U32	Accountability Identifier for this event
SCI0TYPE	U8	Packet data content type: · 0x01: FPGA pre-pended metadata header · 0x02: TOF HG · 0x04: TOF LG · 0x08: TOF MG · 0x10: Target High Range (THR) CSA · 0x20: Target Low Range (TLR) CSA · 0x40: Ion grid (ION) CSA
SCI0CONT	U8	Channels being downlinked (same decomposition as above)
SCI0SPARE1	U13	Spare
SCI0PACK	U1	Always 0b1, showing data is bit-packed
SCI0FRAG	U1	If 0b1, then this channel data spans multiple packets and this is not the last packet (use SCI0FRAGOFF to reconstruct)
SCI0COMP	U1	Data is compressed if 0b1, otherwise not
SCI0EVTNUM	U16	Event number for this event
SCI0CAT	U8	Category assigned to this event (for most recent processing operation)
SCI0QUAL	U8	Quality factor assigned to this category
SCI0FRAGOFF	U16	Starting offset for this data when reconstructing data that spans multiple packets, in 32-bit "dwords" not bytes. Either 0x000 or multiple of 0x3F0
SCI0VER	U16	Science CSC version number for this header
SCI0TIMECOARSE	U16	Lowest 16 bits of integer seconds since epoch for time of this event
SCI0TIMEFINE	U16	Sub-second time of first event, each DN represents 20usec within current second
SCI0SPARE2	U32	Spare
SCI0SPARE3	U32	Spare
SCI0SPARE4	U32	Spare
Variable length raw science data	Varies	Padded with zeroes for 32-bit alignment, if needed
SYNCSCI0PKT	U16	Synchronization marker, fixed 0x3333
CRCSCI0PKT	U16	CRC CCITT-16